

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# -----
# INPUT IMAGE CREATED INSIDE CODE
# -----
img = np.zeros((300, 400), dtype=np.uint8)

# Create sample image
cv2.rectangle(img, (50, 50), (200, 200), 160, -1)
cv2.circle(img, (300, 150), 70, 220, -1)

# Add text
cv2.putText(img, "IMAGE", (90, 270),
            cv2.FONT_HERSHEY_SIMPLEX, 1, 255, 2)

# -----
# HIGH-BOOST MASK
# -----
A = 2

mask = np.array([
    [0, -1, 0],
    [-1, A + 4, -1],
    [0, -1, 0]
], dtype=np.float32)

# Apply high-boost mask
sharpened = cv2.filter2D(img, -1, mask)

# -----
# DISPLAY RESULTS
# -----
plt.figure(figsize=(10, 4))
```

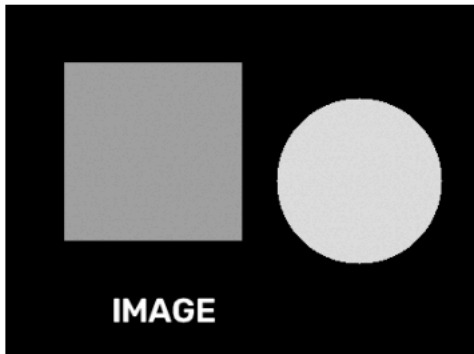
```
plt.subplot(1, 2, 1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(sharpened, cmap='gray')
plt.title("High-Boost Sharpened Image")
plt.axis("off")

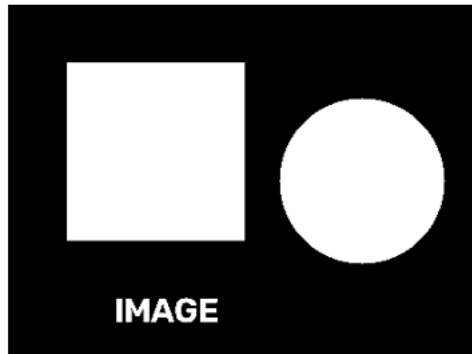
plt.show()

# Display mask
print("High-Boost Mask:")
print(mask)
```

Original Image



High-Boost Sharpened Image



```
High-Boost Mask:
[[ 0. -1.  0.]
 [-1.  6. -1.]
 [ 0. -1.  0.]]
```

